

Common Elevator Failure Modes - Ontario Fleet Analysis

Rocket Elevators - Ontario Fleet Technical Reference

Overview

Based on TSSA incident and inspection data from the Ontario elevator fleet, this document describes the ten most common failure modes, their root causes, and recommended preventive actions. Data reflects 2010-2018 incident reports.

1. Utility and Power Failures

Most frequent cause in the Ontario dataset (> 35% of incidents classified with a root cause). Power interruptions strand passengers and can trigger uncontrolled descent on older DC systems without battery backup.

Prevention: Install UPS for door operator and lighting circuits; verify ARD (Automatic Rescue Device) operation annually.

2. Defective or Failed Components

Second most common category. Includes door operator failures, relay contacts, selector tapes, and position encoders.

Prevention: Implement a predictive maintenance schedule tracking component age against MTBF; replace door operators at 150,000 cycles or 15 years.

3. Door-Related Entrapments

Doors that fail to open at landings are the primary entrapment cause. Causes include worn cam rollers, misaligned door sills, broken vane couplers, and faulty door zone sensors.

Prevention: Clean and lubricate door tracks monthly; test door open/close force and reversal device quarterly.

4. Levelling and Stopping Accuracy

Poor levelling (car floor more than 12 mm from landing sill) is a trip hazard and a TSSA deficiency. Common causes: worn selector tape, incorrect speed profile parameters, or brake adjustment drift.

Prevention: Check levelling at all landings during each maintenance visit; re-calibrate the drive speed profile when levelling error exceeds 6 mm.

5. Rope and Chain Wear

Hoist ropes with > 10 broken wires per lay length, or outer wire wear > 1/3 of original diameter, must be replaced per ASME A17.1 Section 8.6.7.

Prevention: Lubricate ropes with a penetrating lubricant annually; measure rope diameter at multiple points every two years.

6. Hydraulic Leaks and Seal Failures

Oil leaks in the machine room or pit are a fire hazard and an environmental violation. In-ground cylinder failures can contaminate soil and groundwater.

Prevention: Annual oil sampling; install double-bottom cylinders during any cylinder replacement; maintain an oil containment tray under the power unit.

7. Vandalism and Misuse

Approximately 0.5% of Ontario incidents are classified as vandalism, sabotage, or theft. Emergency phones are a frequent target; forced door openings damage interlock mechanisms.

Prevention: Install tamper-resistant emergency phone housings; review CCTV footage after any unexplained fault cluster.

8. Environmental - Weather and Flooding

Flooding from roof drains, burst pipes, and spring thaw accounts for a significant cluster of incidents in the January-March period.

Prevention: Inspect pit drainage pump annually before winter; seal penetrations in the pit wall where water intrusion is a risk.